

Problem Statement and Goals

ProgName

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Table 1: Revision History

Date	Developer(s)	Change
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
...	...	...

1 Problem Statement

1.1 Problem

Around 440,000 Canadians deal with some kind of speech, language, or communication issue [?]; this includes 7-10% of preschool children in Canada [?]. A common speech impediment in children is substitution, which is when an expected phoneme is replaced by another (e.g. a child may pronounce “Rabbit” as “Wabbit”) [?]. For detection of said issues, speech language pathologists (SLPs) and researchers have used various tools for analyzing speech and helping individuals improve. Current phonetic detection tools for researchers and SLPs are proprietary, not compliant with the Personal Health Information Protection Act (PHIPA), and/or poor in performance (e.g. speed and accuracy) [?]. This means that available tools are insufficient for research and for treating substitution speech impediments in children. There should be a substitution treatment tool for child SLPs that is open-source, PHIPA-compliant, performant, and expandable to other professions (e.g. accent coaches) and other speech differences beyond execution.

1.2 Inputs and Outputs

[Characterize the problem in terms of “high level” inputs and outputs. Use abstraction so that you can avoid details. —SS]

1.3 Stakeholders

1.4 Environment

Hardware:

- PC (desktop or laptop, Specs to be determined)
- Microphone (built-in or external)

Software/Firmware:

- Web browser (Google Chrome or Edge)
- Windows 10 and 11 Operating System
- Any required drivers for microphone
- Supports HTTPS connections

2 Goals

3 Stretch Goals

[Goals you hope to get to, but not necessary. They are also helpful if the scope of the original project needs to be expanded. —SS]

4 Custom Documents (CDs)

[For CAS 741: State whether the project is a research project. This designation, with the approval (or request) of the instructor, can be modified over the course of the term. —SS]

[For SE Capstone: List your custom documents (CDs). Potential CDs include usability testing, code walkthroughs, user documentation, formal proof, GenderMag personas, Design Thinking, etc. (The full list is on the course outline and in Lecture 02.) Normally the number of CDs will be two (one per term). Approval of the CDs will be part of the discussion with the instructor for approving the project. The CDs, with the approval (or request) of the instructor, can be modified over the course of the term. —SS]

Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?